

Three Kinds of Sameness: A Complete Classification of Unification Claims

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Abstract The term “unified field theory” is used for three logically distinct assertions: the conversion type (one field physically transforms into another), the geometric type (multiple fields are projections of higher-dimensional geometry), and the algebraic type (multiple theories share a deeper algebraic structure). This paper proves that these three exhaust all testable assertions of the form “physical phenomena A and B are the same thing”—sameness claims come in exactly three forms: map, common projection, shared invariant. The proof uses the formal framework of theories (theories as categories, Halvorson 2019) and the formal duality-equivalence literature (De Haro-Butterfield 2018; Weatherall 2018) as its language. Each of the three forms corresponds to a distinct evidence regime (quantitative adjudication / window narrowing / ontological silence), which explains why the three unifications fail in different ways and require different adjudication experiments—this is the formal basis of the adjudication-matrix methodology. The paper includes an explicit falsification clause: if a fourth sameness structure is found, the classification is revised accordingly.**Keywords** unified field theory; theoretical equivalence; category theory; duality; scientific realism

Introduction

The historical narrative of unified field theory has conflated three different assertions into one word, producing a century of confusion. Distinguishing them (see the companion paper **Three Unifications**) is the first step; proving that they **exhaust** the possibilities is the second—the task of this paper. A completeness proof for a classification elevates it from “one viewpoint among many” to “an enumeration of all possibilities”—the qualification required for that classification to serve as a methodological foundation.

Formal Framework

Theories as categories

We adopt the Halvorson (2019) framework: a physical theory T is a category, whose objects are models of physical systems and whose morphisms are the physically allowed transformations. The “sameness” of two theories then has three basic forms—the standard vocabulary of category theory:

- **Map:** a functor $f : T_A \rightarrow T_B$ (possibly an equivalence), translating A’s structure into B’s structure;
- **Common projection:** there exists a third theory T_C with functors $p_A : T_C \rightarrow T_A$, $p_B : T_C \rightarrow T_B$ (A and B are both quotients/fibers of C);
- **Shared invariant:** there exists an algebraic/categorical invariant K such that both A and B are representations or realizations of K .

The formal duality-equivalence literature (De Haro-Butterfield; Weatherall) chiefly treats the first form (equivalence maps between theories); its framework extends naturally to the latter two.

Definition of a sameness claim

Definition. A “sameness claim” is a proposition $P(A, B)$ supported by observational data: “physical phenomena A and B are the same thing,” such that the realization of P must be a computable structure—a concrete relation between the structures of A and B checkable against data.

Theorem: Completeness of the Trichotomy

Theorem (Completeness of the trichotomy of sameness claims). Let $P(A, B)$ be a sameness claim as defined above. Then the structural basis of P must be one of the following three (or a composite):

- (1) **Map type:** there exists a map preserving A’s observable structure into B (symmetry, duality, or dynamical conversion)—the conversion unification;
- (2) **Projection type:** A and B are both projections/restrictions of a common parent structure C—the geometric unification;
- (3) **Invariant type:** A and B share an algebraic or kinematic invariant K—the algebraic unification.

A fourth structural basis does not exist.

Structure	Physical prototype	Evidence regime	Adjudication mode
Map	Electromagnetic duality (self-dual sector); Lorentz transformations	Quantitatively computable	Direct computation (Schwinger inversion)
Projection	Kaluza-Klein fibration	Window narrowing	Orbital constraints + nuclear-clock window
Invariant	The kinematic algebra of the double copy	Ontological silence	Not adjudicable by computation

Table 1: Table 1: The one-to-one correspondence of three structures, three evidence regimes, and three adjudication modes.

Sketch of proof. The structural basis of a sameness claim is given by the categorical relation between A and B. A basic fact of category theory: any relation between two categories decomposes into composites of three kinds of ingredients—functors (maps), universal constructions (limits/colimits—projections from common domains), and invariants (kernels of representation/module structures). If the realization of P is computable (by definition), its categorical realization must be a constructible functorial relation, hence a composite of the three ingredients. The three ingredients correspond exactly to the three unifications. If a fourth kind, indecomposable into the three, existed, then the realization of P would be non-computable, contradicting the definition. QED (a fully formalized version is left to future work; the present paper’s role is to give the structure of the proof and its falsification clause).

Physical Correspondence: Three Structures $\leftrightarrow$ Three Evidence Regimes

Each structure determines an evidence regime, explaining the different fates of the three unifications:

This correspondence is the formal basis of the adjudication-matrix methodology: each matrix row is first classified as some sameness claim, and the adjudication channel is then configured according to that class’s evidence regime—not a methodological choice, but a structural necessity.

Falsification Clause

The theorem includes an explicit falsification clause: **if there exists a testable sameness claim whose structural basis cannot be decomposed into the map/projection/invariant three, the theorem is falsified and the classification revised.** The historical candidate for a “fourth”—duality from common boundary conditions (bulk-boundary correspondence)—falls within the projection type in this framework (AdS/CFT is a holographic projection); should future physics provide a genuine fourth, the classification extends accordingly. The value of a completeness proof lies precisely in making such revision precisely describable.

Conclusion

The three unifications are not one viewpoint among many but the three possible structures of sameness claims. The completeness makes the differing evidence regimes of the three classes a structural necessity, and the adjudication matrix the operationalization of that structure. This is a first version; formalization details of the proof (the full statement of functorial decomposition) are left to future work; the falsification clause secures its scientific standing.

References

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Statement The categorical language of this paper serves only as a classification skeleton; all physical content comes from the companion papers and the literature.